Math 208, Midterm 2 Name: _____

Signature: _____

Student ID #: _____ Section #: I

- You are allowed a Ti-30x IIS Calculator and one 8.5×11 inch paper with handwritten notes on both sides. Other calculators, electronic devices (e.g. cell phones, laptops, etc.), notes, and books are **not** allowed.
- Some questions require you to explain answers: no explanation, no credit.
- Try to show your work on all questions: no work, no partial credit.
- You may use the back of the exam for scratch work: please submit any additional paper you use.
- Place a box around your answer to each question.
- Raise your hand if you have a question.

1	/10
2	/10
3	/10
4	/10
5	/10
T	/50

Good Luck!

(1) Determine bases for the rowspace (3pts), nullspace (3pts), and column space (3pts) of $A = \begin{pmatrix} 2 & 3 & 5 \\ 8 & 13 & -4 \end{pmatrix}$.

(1pt) What is the rank of A?

(2) Consider the matrix $A = \begin{pmatrix} \sqrt{2}/2 & 0 & -\sqrt{2}/2 \\ 3 & 1 & 1 \\ \sqrt{2}/2 & 0 & \sqrt{2}/2 \end{pmatrix}$.

(a) (5pts) Calculate $\det(A)$

(b) (5pts) Determine whether or not A^{-1} exists, and if so compute it.

(3) Consider the matrix

$$R = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{pmatrix}$$

(a) (5pts) Calculate the matrix product $R^T R$.

(b) (5pts) What is the general solution to the following 3×3 linear system?

$$R\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(4) (10pts) Let R be the matrix from the Problem 3. Find a *nonzero* vector $v \in \mathbb{R}^3$ such that $Rv = v$.

(5) Let $X = \{(x, y) \in \mathbb{R}^2 \mid y = |x| \text{ and } x \geq 0\}$ (here $| \bullet |$ denotes the usual absolute value of a real number $\bullet$.)
a) (5pts) Is X a subspace of $\mathbb{R}^2$? Explain.

b) (5pts) What is the smallest subspace of $\mathbb{R}^2$ containing X?